VT-447V(I)

PROCESS GUIDE

Laminate/Prepreg

UL Approval: E214381 Version: 20/09/2026

Precautions in Handling

- Please be careful when single ply of 1080, 1086, 1078 106,1067,1037 or 1027prepreg is designed to the dielectric layer.
- Dimension stability is the same as Standard FR4 material.
- Please check with your oxide vendor to make sure that our material is suitable with your oxide process. We recommend controlling the peel strength with brown oxide copper over 2 Lb/in.
- For unclad or single sided laminates to be used in multilayer, please brush unclad sides before use.

Storage Condition & Shelf Life

		Prepreg		Laminate
Storage Condition	Temperature	Below 23°C (73°F)	Below 5°C (41°F)	Room
	Relative Humidity	Below 55%	/	/
Shelf Life		3 Months	6 Months	24 Months (Airproof)

- The prepreg exceeding shelf time should be retested.
- Take care in handling thin core laminates as they are easily damaged.
- If the prepreg is not consumed within 48hrs after opening the vacuum package, it is recommended that the bags be resealed.
- Material is available in both long and short grain. The grain direction is indicated on the label with an arrow.

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Prepreg Availability

Dk values are for impedance design

Glass Type	Resin Content	Press Thickness		Dk				Df			
		mil	μm	@ 1GHz	@ 2GHz	@ 5GHz	@ 10GHz	@ 1GHz	@ 2GHz	@ 5GHz	@ 10GHz
1027	71%	1.5	38	3.85	3.75	3.70	3.70	0.014	0.015	0.015	0.016
1027	73%	1.6	41	3.80	3.70	3.65	3.65	0.014	0.015	0.015	0.016
1027	75%	1.8	46	3.75	3.65	3.60	3.60	0.015	0.016	0.016	0.017
1037	73%	2.0	51	3.80	3.70	3.65	3.65	0.014	0.015	0.015	0.016
1037	75%	2.2	56	3.75	3.65	3.60	3.60	0.015	0.016	0.016	0.017
1037	77%	2.4	61	3.70	3.60	3.55	3.55	0.015	0.016	0.016	0.017
1037	79%	2.6	66	3.67	3.57	3.52	3.52	0.015	0.016	0.016	0.017
106	71%	1.9	48	3.85	3.75	3.70	3.70	0.014	0.015	0.015	0.016
106	73%	2.1	53	3.80	3.70	3.65	3.65	0.014	0.015	0.015	0.016
106	75%	2.3	58	3.75	3.65	3.60	3.60	0.014	0.015	0.015	0.016
1067	70%	2.2	56	3.88	3.78	3.73	3.73	0.014	0.015	0.015	0.016
1067	72%	2.4	61	3.83	3.73	3.68	3.68	0.014	0.015	0.015	0.016
1067	74%	2.6	66	3.78	3.68	3.63	3.63	0.015	0.016	0.016	0.017
1067	76%	2.8	71	3.72	3.62	3.57	3.57	0.015	0.016	0.016	0.017
1067	78%	3.0	76	3.69	3.59	3.54	3.54	0.015	0.016	0.016	0.017
1078	67%	3.0	76	3.95	3.85	3.80	3.80	0.014	0.015	0.015	0.016
1078	69%	3.2	81	3.90	3.80	3.75	3.75	0.014	0.015	0.015	0.016
1080	64%	2.8	71	4.03	3.93	3.88	3.88	0.014	0.015	0.015	0.016
1080	66%	3.0	76	3.98	3.88	3.83	3.83	0.014	0.015	0.015	0.016
1080	68%	3.2	81	3.93	3.83	3.78	3.78	0.014	0.015	0.015	0.016
1080	71%	3.5	89	3.85	3.75	3.70	3.70	0.014	0.015	0.015	0.016
1086	67%	3.5	89	3.95	3.85	3.80	3.80	0.014	0.015	0.015	0.016
1086	69%	3.7	94	3.90	3.80	3.75	3.75	0.014	0.015	0.015	0.016
2113	57%	3.9	99	4.21	4.10	4.05	4.05	0.013	0.014	0.014	0.015
2113	59%	4.25	108	4.17	4.07	4.02	4.02	0.013	0.014	0.014	0.015
3313	59%	4.25	108	4.17	4.07	4.02	4.02	0.013	0.014	0.014	0.015
3313	62%	4.65	118	4.08	3.98	3.93	3.93	0.014	0.015	0.015	0.016
2116	53%	4.5	114	4.30	4.20	4.15	4.15	0.013	0.014	0.014	0.015
2116	55%	4.7	119	4.25	4.15	4.10	4.10	0.013	0.014	0.014	0.015
2116	57%	5	127	4.21	4.10	4.05	4.05	0.013	0.014	0.014	0.015
2116	59%	5.2	132	4.17	4.07	4.02	4.02	0.013	0.014	0.014	0.015
2116	61%	5.4	137	4.10	4.00	3.95	3.95	0.014	0.015	0.015	0.016
7628	45%	7.6	193	4.52	4.42	4.37	4.37	0.012	0.013	0.013	0.014
7628	48%	7.8	198	4.46	4.36	4.31	4.31	0.012	0.013	0.013	0.014
7628	51%	8.3	211	4.35	4.25	4.20	4.20	0.013	0.014	0.014	0.015

Remark:

1. Press thickness test condition---Prepreg lamination size 18"*24", Copper Foil---1oz/1oz, Flow---about 5%;
2. Make laminated prepreg to micro-section and measure the thickness with microscope;
   This thickness is used for resistance design calculation.
3. The thickness measured with micrometer is 0.2~0.4 mil larger than that measured with micro-section;
   and mainly used for total thickness design calculation.

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Laminate Availability

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Core Thickness		Stack-up	Resin Content	Dk				Df			
inches	mm			@ 1GHz	@ 2GHz	@ 5GHz	@ 10GHz	@ 1GHz	@ 2GHz	@ 5GHz	@ 10GHz
0.0020	0.051	1067x1	68%	3.93	3.83	3.78	3.78	0.014	0.015	0.015	0.016
0.0025	0.064	1078x1	58%	4.19	4.09	4.04	4.04	0.013	0.014	0.014	0.015
0.0030	0.076	*1080x1	64%	4.03	3.93	3.88	3.88	0.014	0.015	0.015	0.016
0.0030	0.076	1086x1	63%	4.05	3.95	3.90	3.90	0.014	0.015	0.015	0.016
0.0035	0.089	2113x1	54%	4.28	4.18	4.13	4.13	0.014	0.015	0.015	0.016
0.0035	0.089	3313x1	54%	4.28	4.18	4.13	4.13	0.014	0.015	0.015	0.016
0.0040	0.102	2116x1	46%	4.50	4.13	4.08	4.08	0.012	0.013	0.013	0.014
0.0040	0.102	1067x2	68%	3.93	3.83	3.78	3.78	0.014	0.015	0.015	0.016
0.0050	0.127	2116x1	56%	4.23	4.13	4.08	4.08	0.014	0.015	0.015	0.016
0.0050	0.127	1078x2	58%	4.19	4.09	4.04	4.04	0.013	0.014	0.014	0.015
0.0060	0.152	1506x1	45%	4.52	4.42	4.37	4.37	0.012	0.013	0.013	0.014
0.0060	0.152	1086x2	64%	4.03	3.93	3.88	3.88	0.014	0.015	0.015	0.016
0.0070	0.178	7628x1	42%	4.61	4.51	4.46	4.46	0.011	0.012	0.012	0.013
0.0080	0.203	7628x1	44%	4.55	4.45	4.40	4.40	0.012	0.013	0.013	0.014
0.0080	0.203	2116x2	46%	4.50	4.40	4.35	4.35	0.012	0.013	0.013	0.014
0.0080	0.203	3313x2	57%	4.21	4.10	4.05	4.05	0.013	0.014	0.014	0.015
0.0100	0.254	2116x2	56%	4.23	4.13	4.08	4.08	0.013	0.014	0.014	0.015
0.0120	0.305	1506x2	45%	4.52	4.42	4.37	4.37	0.012	0.013	0.013	0.014
0.0140	0.356	7628x2	42%	4.61	4.51	4.46	4.46	0.011	0.012	0.012	0.013
0.0150	0.381	7628x2	43%	4.58	4.48	4.43	4.43	0.011	0.012	0.012	0.013
0.0150	0.381	2116x3	55%	4.25	4.15	4.10	4.10	0.013	0.014	0.014	0.015
0.0210	0.533	7628x3	42%	4.61	4.51	4.46	4.46	0.011	0.012	0.012	0.013
0.0240	0.610	7628x3	42%	4.61	4.51	4.46	4.46	0.011	0.012	0.012	0.013
0.0280	0.711	7628x4	42%	4.61	4.51	4.46	4.46	0.011	0.012	0.012	0.013
0.0310	0.787	7628x4	44%	4.55	4.45	4.4	4.4	0.011	0.012	0.012	0.013
0.0360	0.914	7628x5	43%	4.58	4.48	4.43	4.43	0.011	0.012	0.012	0.013
0.0450	1.143	7628x6	43%	4.58	4.48	4.43	4.43	0.011	0.012	0.012	0.013
0.0490	1.245	7628x7	42%	4.61	4.51	4.46	4.46	0.011	0.012	0.012	0.013
0.0560	1.422	7628x8	42%	4.61	4.51	4.46	4.46	0.011	0.012	0.012	0.013
0.0590	1.499	7628x8	44%	4.55	4.45	4.4	4.4	0.011	0.012	0.012	0.013
0.0660	1.676	7628x9	44%	4.55	4.45	4.4	4.4	0.011	0.012	0.012	0.013

- Thickness: 0.002" (0.05mm) to 0.200" (5mm), available in sheet or panel form.
- Copper Foil: 1/4 to 12oz, HTE or RTF or DST.

Remark: * Standard

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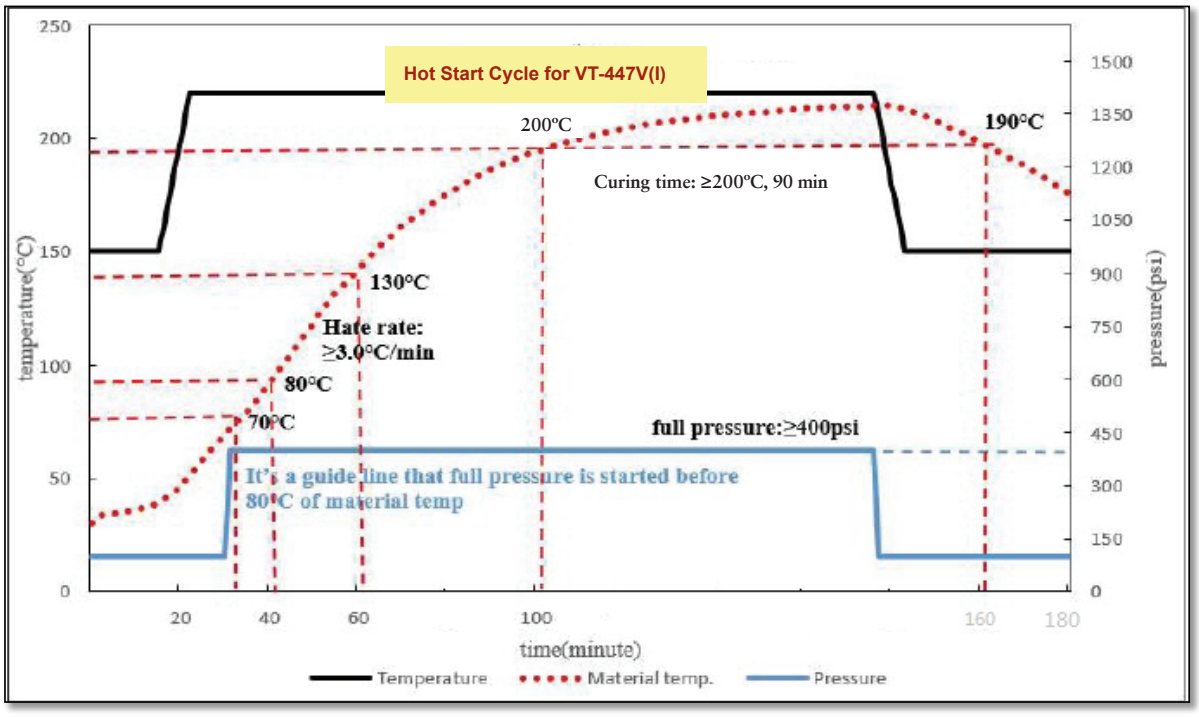
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Press Condition

1. Heating rate (Rate of Rise) of material [Material Temperature]: Programmable Press: : ≥3.0°C/min
2. Curing Temperature & Time: >90min at more than 200°C and peak temperature>210 °C
3. Full Pressure: ≥400psi (28Kg/cm2) should be applied full pressure before 80 °C
4. Vacuuming should be continued until over 140°C [Material Temperature]
5. Cushion for pressure evenness is needed.



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Typical Drilling Conditions

- Drilling parameters should be adjusted depending on hole size, layer count, panel thickness, stack count and stack height etc.
- Please adjust drilling parameters after checking qualities of through holes.
- Suggest Drilling parameter as blow:

Diameter (mm)	Spindle Speed (KRPM)	Feed Rate (mm/s)	Chip Load (μm/rev)	Hit Count
0.25	125	25	10~15	500
1.0	53	31	30~45	800

Desmear Process

- Please test desmear rate and check whether the smear is cleaned clearly by SEM, Ventec could provide the specimen for pilot.
- 1 cycle Plasma and 1 cycle desmear or 2 cycles desmear is recommended.
- Holding time from desmear to PTH process: < 2 hours.

Packaging and baking recommendation

- It is recommended to bake the board before packaging at 125°C/4~8h to avoid moisture causing a decrease in heat resistance.
- If the PCBs needs to be stored for a long time before use, it is recommended to use aluminum foil vacuum packaging.
- If exceed 3 months after packaging , It is best to bake the PCBs at 125°C/4~6h before assembly before use.